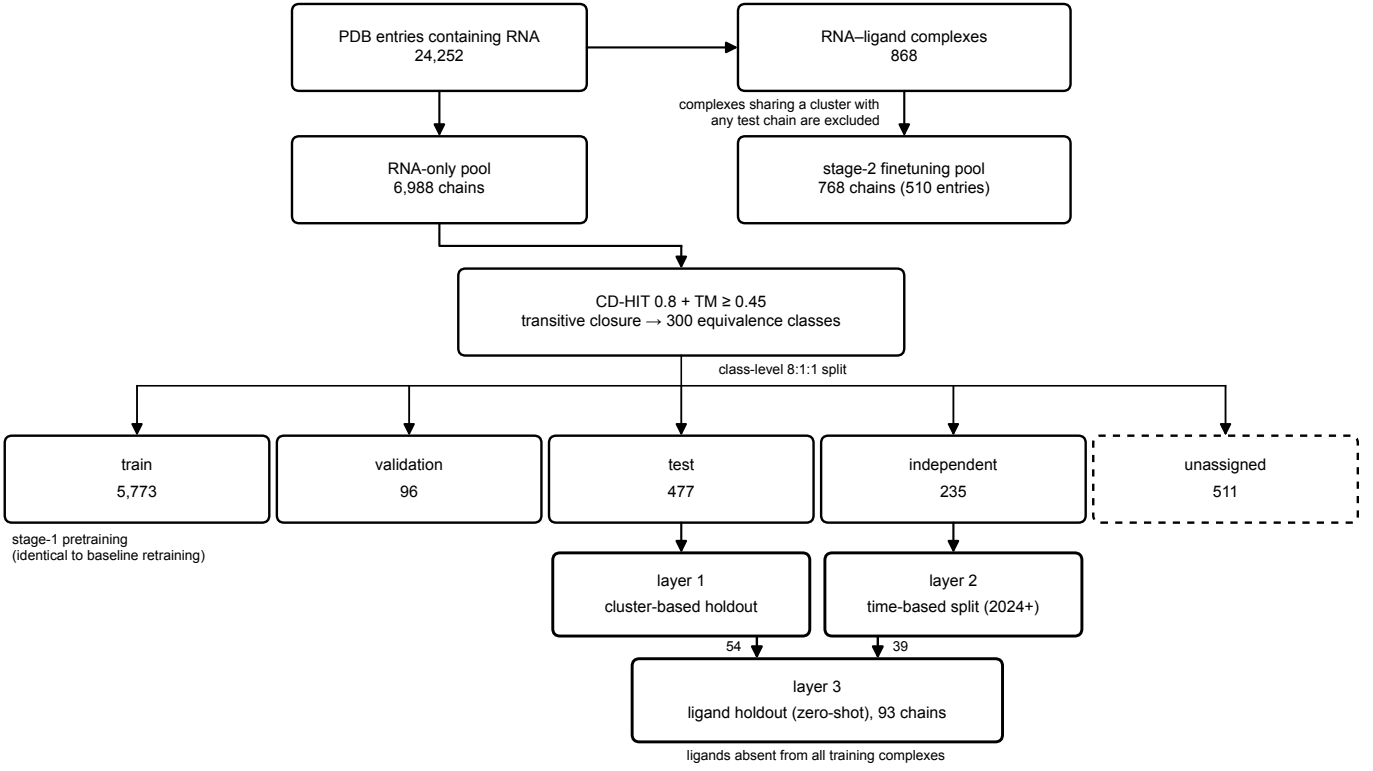
PDB entries containing RNA
24,252
RNA–ligand complexes
868
complexes sharing a cluster with
any test chain are excluded
RNA-only pool
6,988 chains
stage-2 finetuning pool
768 chains (510 entries)
CD-HIT 0.8 + TM ≥ 0.45
transitive closure → 300 equivalence classes
class-level 8:1:1 split
train
5,773
validation
96
test
477
independent
235
unassigned
511
stage-1 pretraining
(identical to baseline retraining)
layer 1
cluster-based holdout
layer 2
time-based split (2024+)
54
39
layer 3
ligand holdout (zero-shot), 93 chains
ligands absent from all training complexes